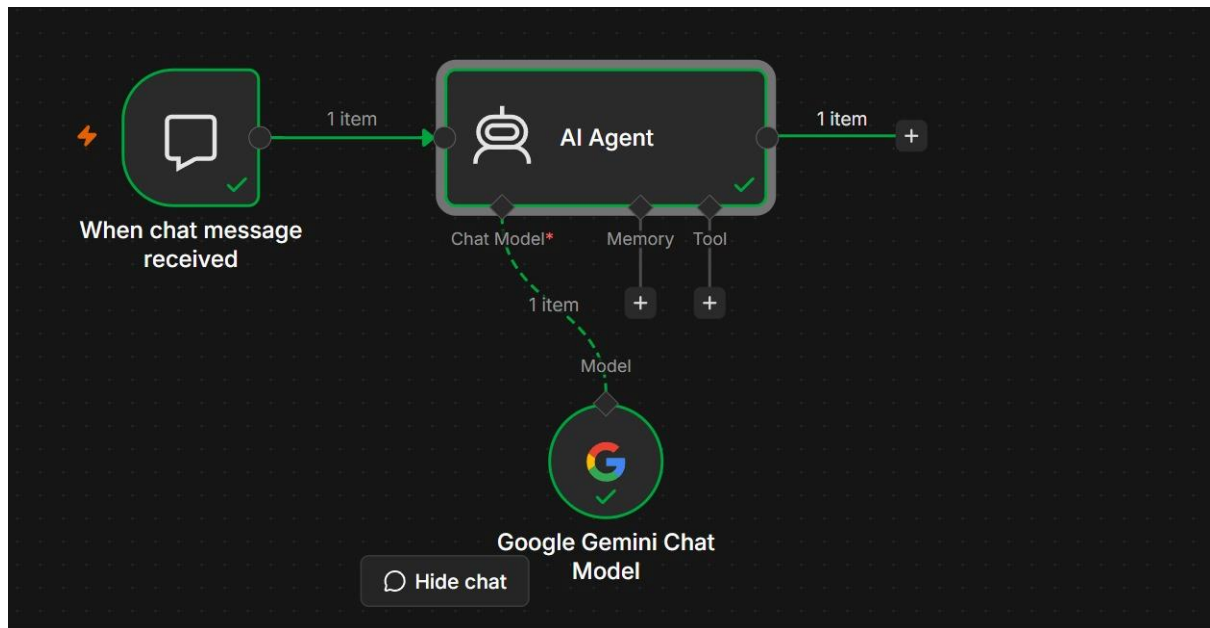


PROJECT-24

AI Flashcard Generator



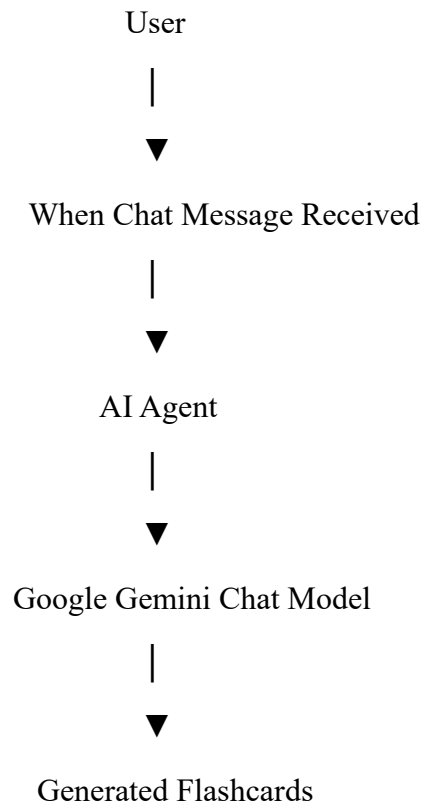
Project Objective

The objective of this project is to build an **AI-powered Flashcard Generator** that automatically creates interactive flashcards from a topic, paragraph, article, lesson, or study material. The AI Agent extracts the most important concepts and converts them into question-and-answer flashcards, making learning faster, easier, and more effective.

The AI Agent can generate:

- Subject-wise flashcards
 - Technical flashcards
 - Language learning flashcards
 - Interview preparation flashcards
 - Exam revision flashcards
 - Programming flashcards
 - Science flashcards
 - History flashcards
 - Medical flashcards
 - General knowledge flashcards
-

Workflow Architecture



Workflow Components

Step	Node	Purpose
1	When Chat Message Received	Receives the user's study topic or learning material
2	AI Agent	Understands the topic and prepares the flashcard generation prompt
3	Google Gemini Chat Model	Generates question-and-answer flashcards

Prerequisites

Before building this project, ensure you have:

- n8n (Cloud or Self-hosted)
- Google AI Studio API Key
- Google Gemini Chat Model configured

- Internet connection

Step 1 – Create Workflow

Open **n8n**

Click

New Workflow

Rename it

AI Flashcard Generator

Step 2 – Add Chat Trigger

Add

When Chat Message Received

Purpose

Receives the user's study topic or learning material.

Example Input

Create 10 flashcards about Machine Learning for beginners.

Output

```
{  
  "chatInput": "Create 10 flashcards about Machine Learning for beginners."  
}
```

Step 3 – Add AI Agent

Connect

When Chat Message Received

|



AI Agent

Configure

Source for Prompt

Connected Chat Trigger Node

Prompt

```
{{ $json.chatInput }}
```

Step 4 – Configure the System Prompt

Paste the following into the **System Message**.

You are an AI Flashcard Generator.

Your task is to create clear, educational, and easy-to-understand flashcards based on the user's topic or study material.

Instructions:

- Understand the user's topic or content.
 - Generate the requested number of flashcards.
 - Each flashcard should contain:
 - Question
 - Answer
 - Keep answers short, accurate, and easy to remember.
 - Focus on the most important concepts and definitions.
 - Use simple language suitable for the intended audience.
 - Avoid duplicate flashcards.
 - Return only the flashcards.
-

Step 5 – Add Google Gemini Chat Model

Click

+ **Chat Model**

Choose

Google Gemini Chat Model

Configuration

Setting	Value
Provider	Google Gemini
Model	Gemini 2.5 Flash (or another available Gemini model)
Authentication	Google AI Studio API Key

Connect it to the **AI Agent**.

Final Workflow

When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

Workflow Execution

Step 1

User enters

Create 10 flashcards about Artificial Intelligence.



Step 2

The **When Chat Message Received** node receives the user's request.



Step 3

The **AI Agent** analyzes the topic and determines the important concepts that should be included in the flashcards.



Step 4

The prompt is sent to the **Google Gemini Chat Model**.

↓

Step 5

Gemini generates educational flashcards containing concise questions and answers.

↓

Step 6

The completed flashcards are returned to the user.

Sample Inputs and Outputs

Example 1

Input

Create 5 flashcards about Artificial Intelligence.

Output

Flashcard 1

Question: What does AI stand for?

Answer: Artificial Intelligence.

Flashcard 2

Question: What is Machine Learning?

Answer: A branch of AI that enables computers to learn from data.

Flashcard 3

Question: Give one real-world application of AI.

Answer: Chatbots and virtual assistants.

Flashcard 4

Question: What is Deep Learning?

Answer: A subset of Machine Learning that uses neural networks.

Flashcard 5

Question: Why is AI important?

Answer: It automates tasks and helps solve complex problems.

Example 2

Input

Generate flashcards for Python programming.

Output

Flashcard 1

Question: What symbol is used for comments in Python?

Answer: #

Flashcard 2

Question: What function is used to display output?

Answer: print()

Flashcard 3

Question: Which keyword defines a function?

Answer: def

Example 3

Input

Create flashcards about World War II.

Output

Flashcard 1

Question: When did World War II begin?

Answer: 1939.

Flashcard 2

Question: When did World War II end?

Answer: 1945.

Flashcard 3

Question: Which organization was formed after World War II?

Answer: The United Nations.

Example 4

Input

Generate biology flashcards about the human heart.

Output

Flashcard 1

Question: How many chambers does the human heart have?

Answer: Four.

Flashcard 2

Question: Which chamber pumps oxygen-rich blood to the body?

Answer: The left ventricle.

Flashcard 3

Question: What is the primary function of the heart?

Answer: To pump blood throughout the body.

Workflow Diagram

User



When Chat Message Received



AI Agent



Google Gemini Chat Model



Generated Flashcards

Error Handling

Error: No Prompt Specified

Cause

The AI Agent did not receive the user's request.

Solution

Verify that the **Prompt** field contains:

```
{{ $json.chatInput }}
```

Error: Authentication Failed

Cause

Invalid Google Gemini API Key.

Solution

Reconnect or update your Google AI Studio credentials.

Error: Too Many Requests

Cause

Gemini API rate limit exceeded.

Solution

- Wait a few minutes.
 - Check your API quota.
 - Retry the request.
-

Learning Outcomes

After completing this project, you will be able to:

- Build an AI-powered Flashcard Generator in n8n.
 - Configure an AI Agent with Google Gemini.
 - Design prompts for educational content generation.
 - Generate study flashcards automatically.
 - Build conversational AI workflows for learning and revision.
-

Real-World Applications

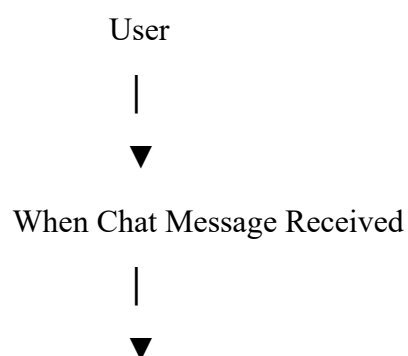
- Student exam preparation
 - Online learning platforms
 - Language learning
 - Technical certification preparation
 - Medical education
 - School and college revision
 - Corporate employee training
 - Competitive exam preparation
 - Programming practice
 - Self-paced learning
-

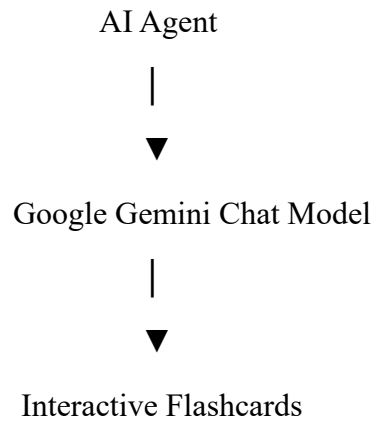
Practice Challenge

Enhance this project by adding:

1. A flashcard count selector (5, 10, 20, or 50 flashcards).
 2. A difficulty level selector (Beginner, Intermediate, Advanced).
 3. A subject selector (Science, History, Programming, Mathematics, etc.).
 4. A Memory node to remember the user's preferred study topics.
 5. A Google Sheets node to save generated flashcards.
 6. An HTTP Request node or Quiz node to convert the generated flashcards into an interactive quiz automatically.
-

Final Workflow





Expected Result

The **AI Flashcard Generator** accepts a topic, lesson, or study material from the user, uses the **Google Gemini Chat Model** to identify the most important concepts, and generates clear, concise, and easy-to-review flashcards in a question-and-answer format. This project teaches prompt engineering, AI-powered educational content generation, and conversational workflow design in n8n, making it an excellent foundation for building intelligent learning, revision, and exam preparation applications.